NOTE 08 - THE DHCP PROCESS: DORA (DISCOVER, OFFER, REQUEST, ACKNOWLEDGE)

DHCP (Dynamic Host Configuration Protocol) gives a device the network settings it needs to join the network automatically, instead of an administrator typing them in on every machine. It is what happens between "the laptop connects to Wi-Fi" and "the laptop can browse".

What DHCP hands out

Setting	Purpose
IP address	The host's own address, taken from a pool (scope)
Subnet mask	Which addresses are local, and therefore where the local network ends
Default gateway	The router used to reach anything outside the local subnet
DNS servers	Which resolvers to query for name resolution
Lease time	How long the client may keep the address before renewing
Optional options	Domain name, NTP servers, PXE boot server, VoIP settings, WPAD proxy

Ports: the server listens on UDP 67, the client listens on UDP 68. DHCPv6 uses UDP 546/547.

DORA - the four steps

D - DISCOVER (client -> broadcast)

The client has no IP address, so it cannot send a normal unicast packet. It broadcasts from `0.0.0.0` to `255.255.255.255:67` with a DHCPDISCOVER message: "Is there a DHCP server on this network?" The message may include the client's MAC address and a list of parameters it would like.

O - OFFER (server -> client)

Every DHCP server that hears the Discover replies with a DHCPOFFER containing a proposed IP address, subnet mask, gateway, DNS servers and lease time. If several servers answer, the client simply picks one offer. The offer is a proposal, not a commitment.

R - REQUEST (client -> broadcast)

The client broadcasts a DHCPREQUEST naming the offer it has chosen. It is broadcast so that the other servers know their offers were declined and can return those addresses to the pool. This is the step where the client formally asks for the address.

A - ACKNOWLEDGE (server -> client)

The chosen server replies with a DHCPACK (a formal Acknowledgement) confirming the address and the full set of options, together with the lease duration. The client now configures its interface and can communicate. If the server refuses (for example the address was taken in the meantime) it sends a DHCPNAK and the client restarts DORA from Discover.

Timing picture

```
Client (no IP)                              DHCP Server (UDP 67)
     |                                             |
     |---- DHCPDISCOVER (broadcast) --------------->|
     |<--- DHCPOFFER  (your IP is 192.168.1.57) ----|
     |---- DHCPREQUEST (I want 192.168.1.57) ------>|
     |<--- DHCPACK  (confirmed, lease 24h) ---------|
     |
  Client configures IP / mask / gateway / DNS and is online
```

Other DHCP messages worth knowing

Message	Direction	Meaning
DHCPDISCOVER	Client -> broadcast	Looking for servers
DHCPOFFER	Server -> client	Proposed configuration
DHCPREQUEST	Client -> broadcast	Accepting one offer / renewing / declining
DHCPACK	Server -> client	Confirmed
DHCPNAK	Server -> client	Refused - start again
DHCPRELEASE	Client -> server	Giving the address back
DHCPINFORM	Client -> server	Client already has an IP and just wants other options
DHCPDECLINE	Client -> server	Address is already in use (conflict)

Lease lifecycle

- Half of the lease time (T1) the client tries to renew directly with its original server.
- At 87.5% (T2) it rebinds, broadcasting to any server that will answer.
- If the lease expires, the address is returned to the pool and the client starts over at Discover.
- A DHCP reservation binds a specific MAC address to a fixed IP, which is how servers and printers get a stable address while still being managed centrally.

Why a SOC analyst cares

- Rogue DHCP servers are a classic attack: an attacker answers Discover with their own offer and sets their IP as the default gateway or DNS server, giving them a man-in-the-middle position (MITRE ATT&CK T1557 - Adversary-in-the-Middle).
- DHCP starvation exhausts the address pool by requesting leases with many fake MAC addresses, so legitimate clients cannot get online (a denial-of-service).
- Troubleshooting: a host that cannot reach anything often has an `169.254.x.x` APIPA address, which means DORA failed and no DHCP server answered.
- Log evidence: DHCP server logs (lease assignments, MAC addresses, hostnames) are what link an IP address back to a device and a user during an investigation - essential because the IP in an alert may have been reassigned since.
- Defences: DHCP snooping on the switch blocks unauthorised DHCP servers, and port security limits MAC flooding.

Key points to remember

- DORA = Discover -> Offer -> Request -> Acknowledge, running over UDP 67/68.
- Discover and Request are broadcast by the client; Offer and Acknowledge come from the server.
- The client ends up with an IP, mask, gateway, DNS servers and a lease.
- DHCP has no built-in authentication, which is why rogue servers, snooping and lease logs matter so much in an investigation.